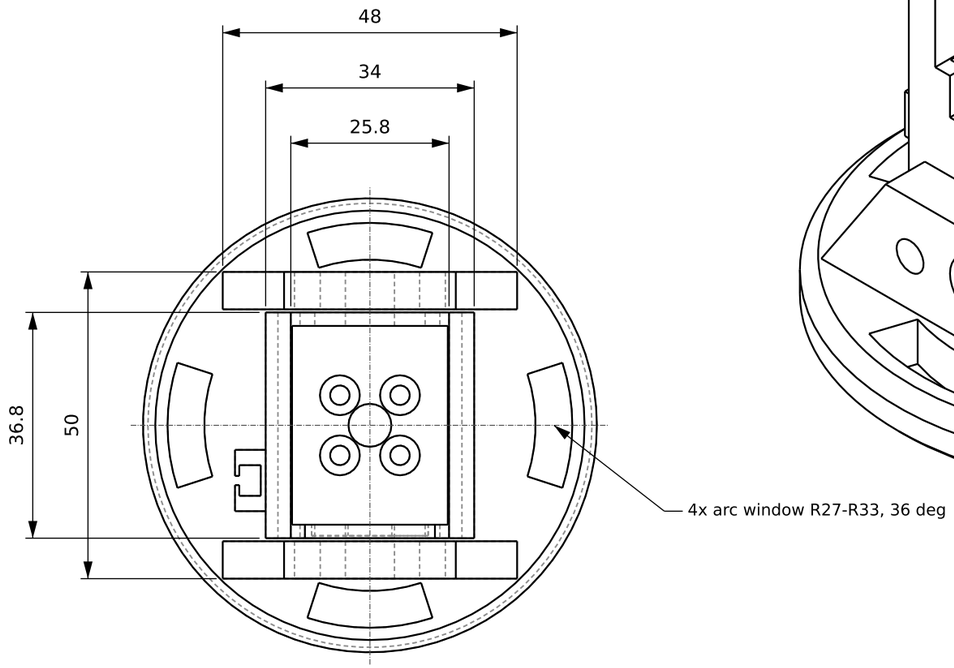
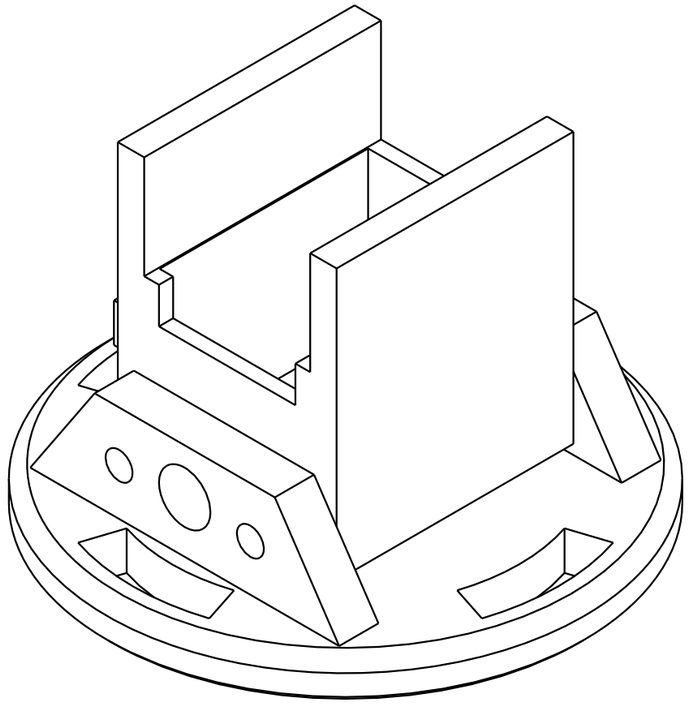
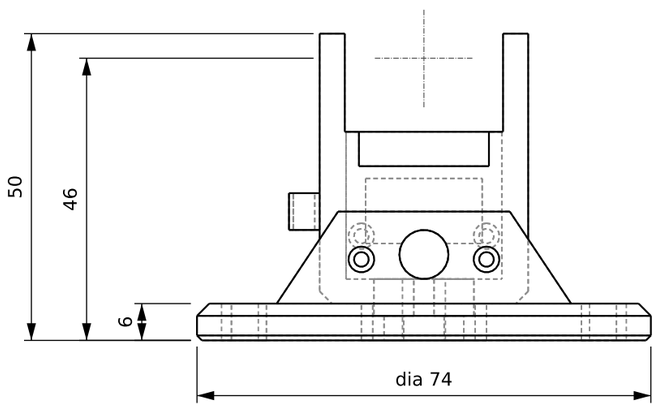


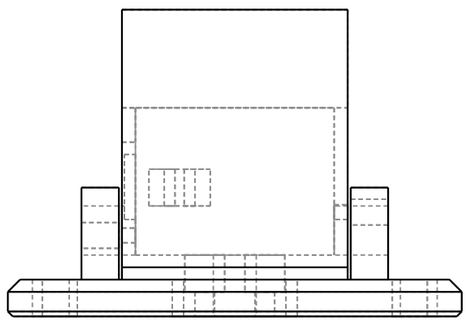
TOP



ISOMETRIC (NOT TO SCALE)



FRONT



RIGHT

FEATURES (in order)

1. Turntable: cylinder dia 74, z 0..6; chamfer 2 top edge, 0.8 bottom edge.
2. Cradle: box x -17..17, y -18.4..18.4, z 5..34; chamfer 3 on the 4 edges parallel to Y.
3. Posts: boxes x -17..-12.9 and 12.9..17, y -18.4..18.4, z 30..50.
4. Cheeks: trapezoid (-24,6) (24,6) (14,21) (-14,21) in XZ, extruded y 18.9..25 and y -25..-18.9.
5. Cable clip (TAL-002) on the back face: centre (-17, -9, 21), cable along Z, face normal -X.
6. Servo pocket P (sheet TAL-001) with the servo frame origin at (0, 0, 46); servo +Z (shaft) -> +Y, servo +X -> +Z (body hangs down), servo +Y -> +X.
7. Servo opening: cut box x -12.7..12.7, y -16.2..16.2, z 10..80.
8. Cable hole: dia 8 along Y, through all, at x = 0, z = 14.
9. Pan horn: centre dia 7, z -1..11. 4x dia 3.2 at (+/-4.9, +/-4.9), z -1..4, counterbore dia 6.5 z 4..11 (M3 x 6 from inside the cradle).
10. Turntable windows: 4x arc slot between R27 and R33, -18..+18 deg about 0/90/180/270 deg, through.
11. Case screw holes (sheet TAL-002): seat 18.5, through the cheeks: shaft face at (x, z) = (+/-10.2, 17), y +15.9..+18.5; connector face at (+/-10.2, 13.2), y -18.5..-15.9. Each is a dia 2.4 hole from the servo face to the seat, then a dia 4.2 counterbore from the seat outward through the part.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.

WORK SET A: base, shoulder, upper arm, hub caps

THIRD ANGLE

PROJECT TALOS 5-axis arm	EST. MASS 39 g
PART **Shoulder (turntable + cradle)**	SCALE 1:1
PART NO. **TAL-102**	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-03
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 5 of 12